

AI Agents Integration Proposal, Team Allocation, and 3-Week Delivery Plan

Prepared for the Preloved Exchange project. This document consolidates the client requirement, proposed implementation scope, role allocation for a 7-person team, and a week-by-week execution plan for Sprint 5.

Sprint Snapshot

Sprint	Sprint 5
Duration	3 weeks
Team Size	7 members
Primary Goal	Add 2 AI agents
Supporting Deliverable	Screen recording demo

Spring Boot Backend

MySQL + MyBatis-Plus

File Upload Support

AI API Integration

Recommendation Logic

Current System Context

Based on the current backend structure, the existing application already includes modules for:

- User authentication and profile management
- Item listing, item filtering, and item detail retrieval
- Category tree and filtering support
- Image/file upload
- Messaging between users
- Announcements and general marketplace operations

This makes Sprint 5 suitable for extending the current marketplace with AI-assisted listing creation and personalized recommendations without redesigning the full system.

1. Client Requirement Summary

The client requested two new AI agents for the website. The first is an AI listing assistant for sellers, which should generate draft listing details after a user uploads a product photo, including title, description, category, condition, and estimated price suggestion. The seller must still review and edit the draft before publishing. The second is a recommendation agent for logged-in users, which should recommend second-hand products based on interests, browsing activity, saved items, or preferred categories. The client also requested a screen recording demo covering the new sprint features and all current app features.

Interpretation for Sprint 5: build two practical AI-assisted workflows that integrate an external AI API with the existing website logic and database, while keeping users in control of final publishing decisions.

2. Sprint Objective

The overall objective of Sprint 5 is to deliver the following outcomes within 3 weeks:

- Implement an AI Listing Assistant workflow for sellers.
- Implement a Recommendation Agent workflow for logged-in users.
- Integrate an external AI API into the current application architecture.
- Store or derive the supporting data needed for recommendation logic.
- Provide a complete demo video of both new features and the current product features.

3. Functional Scope

3.1 AI Agent 1 - Listing Assistant for Sellers

- User uploads one or more product images.
- System sends image context and prompt to external AI API.
- AI returns draft title, description, suggested category, condition, and price range.
- User reviews, edits, accepts, or partially modifies the draft.
- Final listing is saved only after explicit user confirmation.

3.2 AI Agent 2 - Recommendation Agent

- Recommendations are available for logged-in users only.
- Input signals may include interests, preferred categories, browsing history, and saved items.
- Backend combines user behavior data with recommendation rules or AI-assisted ranking.
- Frontend displays personalized products in a dedicated recommendation section.
- Recommended items should link directly to the existing item details flow.

4. Non-Functional Requirements

- Users must remain in control: AI output is draft-only and must be editable.
- The AI integration should not block normal listing creation if the AI service fails.
- Recommendation responses should be reasonably fast and support fallback defaults.
- Sensitive API keys must be stored securely through environment configuration.
- Testing should cover prompt handling, API failures, invalid images, and empty recommendation scenarios.

5. Suggested Technical Approach

5.1 Existing Backend Fit

The current backend already contains controllers and services around items, categories, file upload, users, profiles, and messaging. This supports a practical implementation path:

- Reuse file upload flow for seller product images.
- Add a dedicated AI service layer to call the external AI API.
- Extend item-related services for AI draft generation and recommendation retrieval.
- Store recommendation signals in the database or derive them from existing interaction data.

5.2 Proposed Backend Additions

AREA	SUGGESTED WORK	REASON
AI Integration	Add AI service or provider client for prompt construction, API calls, and response parsing.	Centralizes external AI logic and keeps controllers clean.
Listing Draft Endpoint	Create endpoint such as <code>/api/v1/items/ai-draft</code> for seller-side draft generation.	Separates AI draft generation from final item publishing.
Recommendation Endpoint	Create endpoint such as <code>/api/v1/items/recommendations</code> for logged-in users.	Supports personalized feed without affecting existing filter APIs.
Behavior Data	Track viewed items, saved items, and preferred categories.	Improves recommendation relevance and supports future analytics.
Fallback Strategy	Return default category-based recommendations or empty-safe responses when AI is unavailable.	Prevents total feature failure in demo or production environments.

5.3 Suggested Frontend Additions

- Add AI-assisted listing creation step after image upload.
- Pre-fill listing form with AI suggestions while allowing manual edits.
- Add recommendation panels to home page, dashboard, or user landing area.
- Display loading, retry, and fallback states clearly.

6. Team Allocation for 7 Members

Since the team contains both frontend and backend members and names were not provided, the allocation below uses role placeholders. Replace Member A to Member G with actual names before submission if needed.

MEMBER	ROLE	PRIMARY RESPONSIBILITY	MAIN DELIVERABLES
Member A	Frontend Lead	AI listing assistant UI flow	Upload-to-draft interaction, editable listing form, loading and error states
Member B	Frontend Developer	Recommendation UI and personalization display	Recommendation section, item cards, logged-in user view integration
Member C	Frontend / Integration	Shared UI polish and demo preparation support	Cross-page consistency, API integration fixes, demo script support, fallback UX
Member D	Backend Lead	AI listing assistant backend	External AI API client, prompt design, image-to-draft endpoint, validation logic
Member E	Backend Developer	Recommendation service	User-signal retrieval, recommendation ranking logic, recommendation API
Member F	Backend / Database	Schema changes and behavior tracking	Browsing history or saved-item support, new tables/fields, test data setup
Member G	QA / PM / Full-stack Support	Testing, coordination, documentation, final demo video	Test plan, regression testing, sprint report, screen recording, final merge verification

Recommended coordination model: one frontend lead, one backend lead, one integration-focused teammate, and one quality/documentation owner to reduce merge conflicts and last-minute demo risk.

7. Weekly Plan (3 Weeks)

WEEK	FOCUS	FRONTEND TASKS	BACKEND TASKS	TEAM OUTPUT
Week 1	Planning and foundation setup	Design listing assistant flow, recommendation layout, wireframes, and UI states	Choose AI API, design endpoints, prepare prompt strategy, design DB changes	Confirmed architecture, API contract, UI mockups, task board, risk list
Week 2	Core implementation	Build listing draft interface and recommendation components, connect APIs	Implement AI draft endpoint, recommendation endpoint, behavior tracking, fallback logic	Working end-to-end prototype for both AI features
Week 3	Testing, polishing, and demo	Improve UX, edge cases, loading states, mobile responsiveness, demo-ready screens	Fix bugs, improve accuracy, secure config, optimize queries, finalize test data	Stable sprint release, regression-tested build, screen recording demo, final report

8. Detailed Task Breakdown by Role

Frontend Workstream - Design seller journey from upload to AI-generated draft. - Build editable fields for title, description, category, condition, and price. - Design personalized recommendation modules for logged-in users. - Support loading, empty, retry, and API failure states. - Prepare polished demo screens for screen recording.	Backend Workstream - Connect external AI API for image-based listing draft generation. - Map AI output into internal DTOs and validate suggested values. - Capture or infer user preference signals for recommendation generation. - Return ranked item recommendations through a new API endpoint. - Secure API keys and ensure service fallback behavior.
--	--

9. Expected Deliverables

- FEATURE** AI Listing Assistant for sellers
- FEATURE** Recommendation Agent for logged-in users
- API** External AI API integration in backend
- DATA** Recommendation input data model or tracking logic
- QUALITY** Test cases and regression check results
- DEMO** Screen recording of current app features and Sprint 5 features
- DOCUMENT** Sprint requirement, task allocation, and weekly plan report

10. Risks and Mitigation

RISK	IMPACT	MITIGATION
AI API output is inconsistent	Draft quality may vary	Use strict prompt format, backend validation, and allow full user editing
Recommendation data is incomplete	Recommendations may feel weak at first	Use category preference fallback and saved-item signals as baseline
External API downtime or quota issues	Demo risk and feature interruption	Prepare fallback demo data and non-AI backup response path
Integration delays near deadline	End-to-end flow may break late	Do mid-sprint integration in Week 2 and assign one integration owner
Uneven workload distribution	Some members may block others	Split work by vertical ownership and hold short syncs every 2 to 3 days

11. Demo Recording Recommendation

The requested screen recording can follow this order for a clear presentation:

1. Briefly introduce the application and current marketplace purpose.
2. Show current core features: registration/login, browsing items, filtering categories, viewing item details, messaging, and listing creation.
3. Show AI Listing Assistant: upload product image, receive AI draft, edit details, and publish or save the listing.
4. Show Recommendation Agent: log in as a user, open personalized recommendations, and click through suggested products.
5. Close with a short summary of how AI improved usability for sellers and buyers.

12. Submission Note

This document is intentionally standardized for academic sprint planning. It can be submitted directly as a requirement and allocation report, or lightly customized by replacing placeholder names, adding actual sprint dates, and attaching the final screen recording link when available.

Recommended final customization before submission: replace Member A to Member G with actual names, add your course name and student IDs, and optionally insert the exact AI provider chosen by the team.